

energy storage is required because of the various challenges that renewables have. The Government of India's priority and focus is to promote pump storage, because we are completely dependent on other critical minerals that come from other countries, so that's the way it has gone about. If you see, it's almost like 130 new projects have got registered in the past couple of years in developing pump storages across the country. That's a positive development and the tech has also become much more advanced. Earlier, pump storage was developed in five to six years and now it takes about three to four years. So the focus around pump storage is the right way, till the time we have a permanent solution to go towards a battery solution only.

Q: When do you think the energy generated from a mix of renewable sources will cross the threshold versus conventional sources, which is largely thermal?

RAJ KUMAR ROY: I think, to a really large extent, governments—whether state or federal—need to invest in storage technologies. And by no means I am saying battery. It could be pump or hydrogen because you need somewhere to store something, to supply during 60 per cent of the time. For example, Germany today has this kind of an arrangement. So we are consuming approximately 12 per cent of our energy mix from renewable sources; forget about storage capacity. Germany has gone to 49 per cent, but then what happened? Their thermal plants had started backing down because it was economically not sustainable to keep them running. But the issue is that the moment you go beyond 40, you need somewhere to store that energy, and storage has a cost. **BT**

ROUND-THE-CLOCK RENEWABLE ENERGY: THE NEXT FRONTIER

One of the biggest challenges in the global movement to 100 per cent renewables today is how to balance supply and demand. Another aspect of that challenge is to ensure stable, sustainable and equitable distribution of renewable energy at all hours of the day throughout the year, without the support of fossil fuels or nuclear energy. Is a 100 per cent renewable energy ecosystem even possible, given that India has just scratched the surface of the industry? Stakeholders from the industry share their thoughts on the subject

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Q: Storage has emerged as a big challenge. For the renewable energy sector, what is the bigger ask right now if this is not achieved?

RAHUL WALAWALKAR, PRESIDENT & MD, CES (INDIA), AND PRESIDENT, IESA: On the storage side, we are very happy especially if you talk about the Budget. Many of the industry requests have been accepted by the finance ministry. One was the viability gap funding for gigawatt hours of storage, which will help many of the projects that are coming up this year. There is also immediate reduction of import duties for cells that can possibly be used for both batteries and electrical vehicles. There is one more thing, which is the rationalisation of GST. It was a big pending ask, because many times we keep on talking about reducing the cost by 30 per cent. But right now, if you are doing a new project, you are ultimately paying almost 40-43 per cent in duties and taxes. So that's something that especially for the next two to three years, if we can provide some relief, it will achieve the immediate reduction. And the demand will support the manufacturing because we are going to see a lot of manufacturing being set up and most of the manufacturing facilities that are happening in terms of giga factories, will help serve both stationary as well as e-mobility. But right now, demand generation on e-mobility is happening at a much faster pace. That's where the expectation is in the first set of giga factories, which will mainly be catering to the e-mobility sector. We are now expecting the power ministry to release a follow-